

Math 235 Practice Final Exam

Q1. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 7 \\ 1 & 1 & -3 \end{bmatrix}$

- (a) Find the characteristic polynomial of A .
- (b) Diagonalize A , if possible. If A cannot be diagonalized, explain why.

Q2. Compute the determinant of the following matrix $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \\ 1 & k & k^2 & k^3 \end{bmatrix}$. Your solution

should be a polynomial in k .

(Hint: Recall that $k^2 - 1 = (k - 1)(k + 1)$ and $k^3 - 1 = (k - 1)(k^2 + k + 1)$.)

Q3. Let V be the vector space of continuous functions that are linear combinations of $f(t) = \cos t$, $g(t) = t^2 - 1$, $h(t) = t^2 + 1$, and $r(t) = t^2 + \cos t$.

- (a) Find a basis $\mathcal{B}$ for V .
- (b) Find the $\mathcal{B}$ -coordinates for $p(t) = 3 \cos t + t^2$, where $\mathcal{B}$ is the basis from part (a).

Q4. Consider the matrix $A = \begin{bmatrix} 6 & 6 \\ -3 & 0 \end{bmatrix}$.

- (a) Find the eigenvalues of A .
- (b) Find an invertible matrix P such that $A = PCP^{-1}$ for some rotation-scaling matrix C (i.e., $C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ for some real numbers a, b).

Q5. It is known that the matrix $B = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & 3 \\ 1 & 1 & 1 \end{bmatrix}$ is invertible. Suppose C is a 3×3 matrix such that $BCB = I_3$ (the 3×3 identity matrix). Is C necessarily invertible? If yes, find the inverse of C . If no, explain why.

Q6. Let $\mathbb{R}^{2 \times 2}$ denote the vector space of 2×2 real matrices, and define a transformation

$$S : \mathbb{R}^{2 \times 2} \rightarrow \mathbb{R}^{2 \times 2},$$

$$S(A) = A^T - A,$$

where A^T is the transpose of the matrix A .

- (a) Show that the transformation S is linear.
- (b) Find a basis of the range of S and a basis of the null space of S . Explain how the dimensions of these subspaces are related.

Q7. Let $A = \begin{bmatrix} 1 & 0 & 1 \\ 3 & 2 & 0 \\ 5 & 2 & 2 \\ -2 & -2 & 1 \end{bmatrix}$. Find

- (a) a basis of $Nul(A)$,
- (b) a basis of $Col(A)$,
- (c) a basis of $Row(A)$,
- (d) the rank of A .